Practical 3 — Superstore discounts and profit

Association, group comparisons and business interpretation · 307312 Applied Statistics

- Companion to association and group comparisons in Part 2.
- Complete Practicals 1 and 2.
- Allow two sessions of about 40–50 minutes.

By the end, you should be able to:

- Compare sales, profit and weighted margins.
- Summarize groups using distributions as well as totals.
- Use scatterplots and correlations with the correct observational unit.
- Distinguish a useful business lead from a causal conclusion.

1. The commercial manager's question

> Which product groups deserve a discount and profitability review?

- **Source:** Tableau's fictitious Superstore, downloaded 6 November 2026.
- **Scope:** US orders placed in 2023–2025.
- **Line analysis:** 6,682 product lines for categories and discount bands.
- **Order analysis:** 3,322 orders for basket sales and profit.
- **Units:** sales and profit in dollars; discount as a fraction.
- **Limit:** discount decisions were not randomized; product mix and other factors differ across groups.

2. Import and construct useful variables

```
data_dir <- if (dir.exists("data")) "data" else if (dir.exists(file.path("..", "data")))
        file.path("..", "data") else file.path("..", "..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
lines <- read.csv(file.path(data_dir, "business", "superstore-lines.csv"),
                  stringsAsFactors = FALSE, colClasses = c(postal_code = "character"))
orders <- make_orders(lines)
lines <- lines %>% mutate(
  line_margin_pct = 100 * profit / sales,
  loss_line = profit < 0,
  discount_band = cut(discount, breaks = c(-Inf, 0, .2, Inf),
    labels = c("None", "Above 0 to 20%", "Above 20%"), ordered_result = TRUE))
stopifnot(!anyNA(lines$discount_band), all(lines$sales > 0))
```

- `data/business/superstore-lines.csv`; guide: `data/business/README.md`.
- A 20% discount belongs to the middle band; zero belongs to **None**.
- These bands are analyst-defined categories, not source classifications.
- A line's margin is profit divided by its recorded sales.
- We do not construct a discount-adjusted revenue or refund measure from undocumented accounting assumptions.

3. Compare categories fairly

```
category_results <- lines %>% group_by(category) %>%
  summarise(line_count = n(), total_sales = sum(sales), total_profit = sum(profit),
            median_line_profit = median(profit), loss_line_pct = 100 * mean(loss_line),
            .groups = "drop") %>%
  mutate(overall_margin_pct = 100 * total_profit / total_sales)
knitr::kable(category_results, digits = 2)
```

category	line_count	total_sales	total_profit	median_line_profit	loss_line_pct	overall_margin_pct
Furniture	1435	526612.5	15432.88	8.64	33.38	2.93
Office Supplies	4024	472949.9	82754.18	6.94	14.66	17.50
Technology	1223	564423.2	94770.69	26.59	14.88	16.79

```
ggplot(category_results, aes(category, overall_margin_pct)) +
  geom_col(fill = petra_ink) +
  labs(x = NULL, y = "Total profit / total sales (%)")
```



- Furniture has substantial sales but a much thinner overall margin in this extract.
- Category totals describe business contribution; median line profit describes a typical line.
- The loss-line rate uses lines as its denominator.
- A category can contain both profitable and loss-making subcategories.

Why an average of percentages can mislead

```
margin_comparison <- lines %>% group_by(category) %>%
  summarise(equal_line_mean_margin_pct = mean(line_margin_pct),
            sales_weighted_margin_pct = 100 * sum(profit) / sum(sales), .groups = "drop")
knitr::kable(margin_comparison, digits = 2)
```

category	equal_line_mean_margin_pct	sales_weighted_margin_pct
Furniture	3.89	2.93
Office Supplies	14.30	17.50
Technology	15.31	16.79

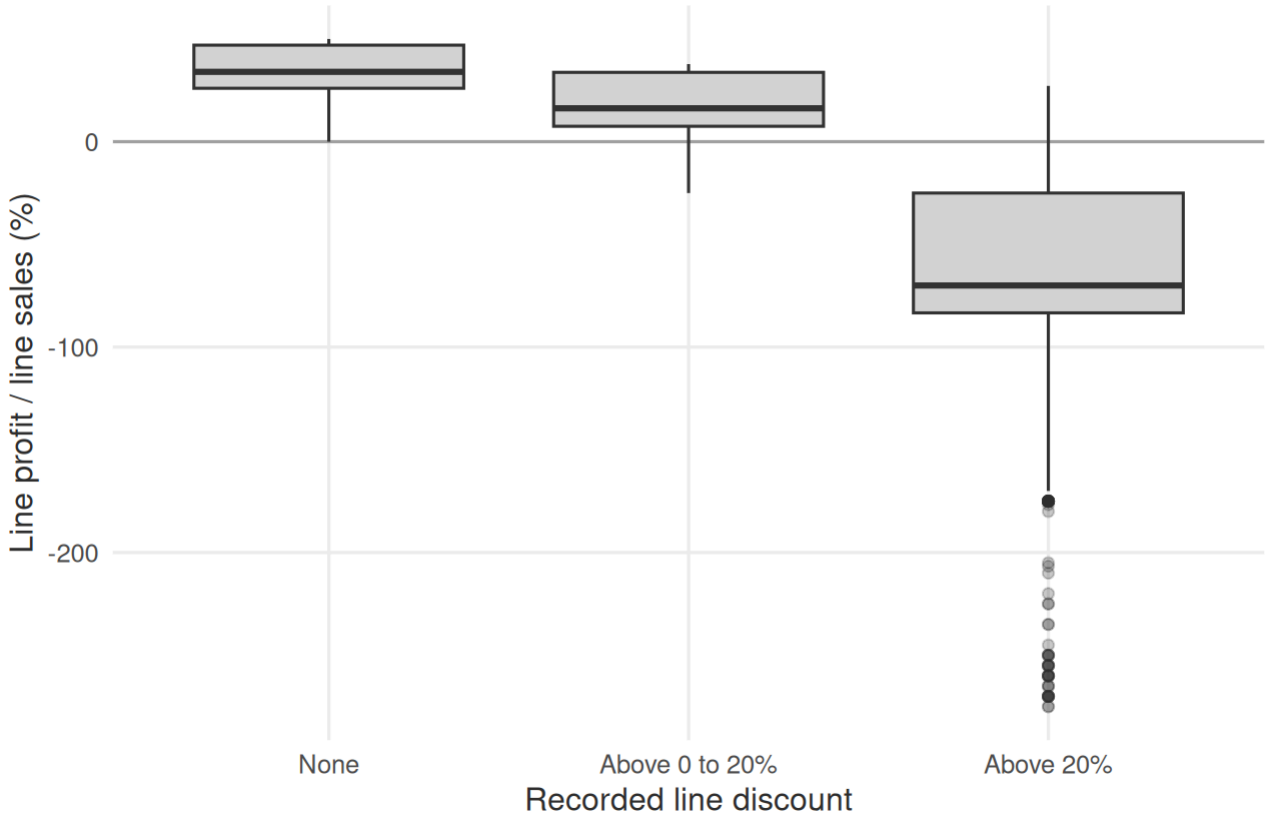
- The first gives every line equal weight.
- The second gives each dollar of sales equal weight.
- Use the second to report the overall category margin.

4. Examine discount groups

```
discount_results <- lines %>% group_by(discount_band) %>%
  summarise(line_count = n(), total_sales = sum(sales), total_profit = sum(profit),
            median_line_profit = median(profit), loss_line_pct = 100 * mean(loss_line),
            .groups = "drop") %>%
  mutate(overall_margin_pct = 100 * total_profit / total_sales)
knitr::kable(discount_results, digits = 2)
```

discount_band	line_count	total_sales	total_profit	median_line_profit	loss_line_pct	overall_margin_pct
None	3208	739090.3	216829.89	15.90	0.00	29.34
Above 0 to 20%	2536	569487.1	63934.02	7.05	13.68	11.23
Above 20%	938	255408.2	-87806.16	-17.86	96.38	-34.38

```
ggplot(lines, aes(discount_band, line_margin_pct)) +
  geom_hline(yintercept = 0, colour = "#A3A3A3") +
  geom_boxplot(fill = petra_accent, outlier.alpha = .25) +
  labs(x = "Recorded line discount", y = "Line profit / line sales (%)")
```



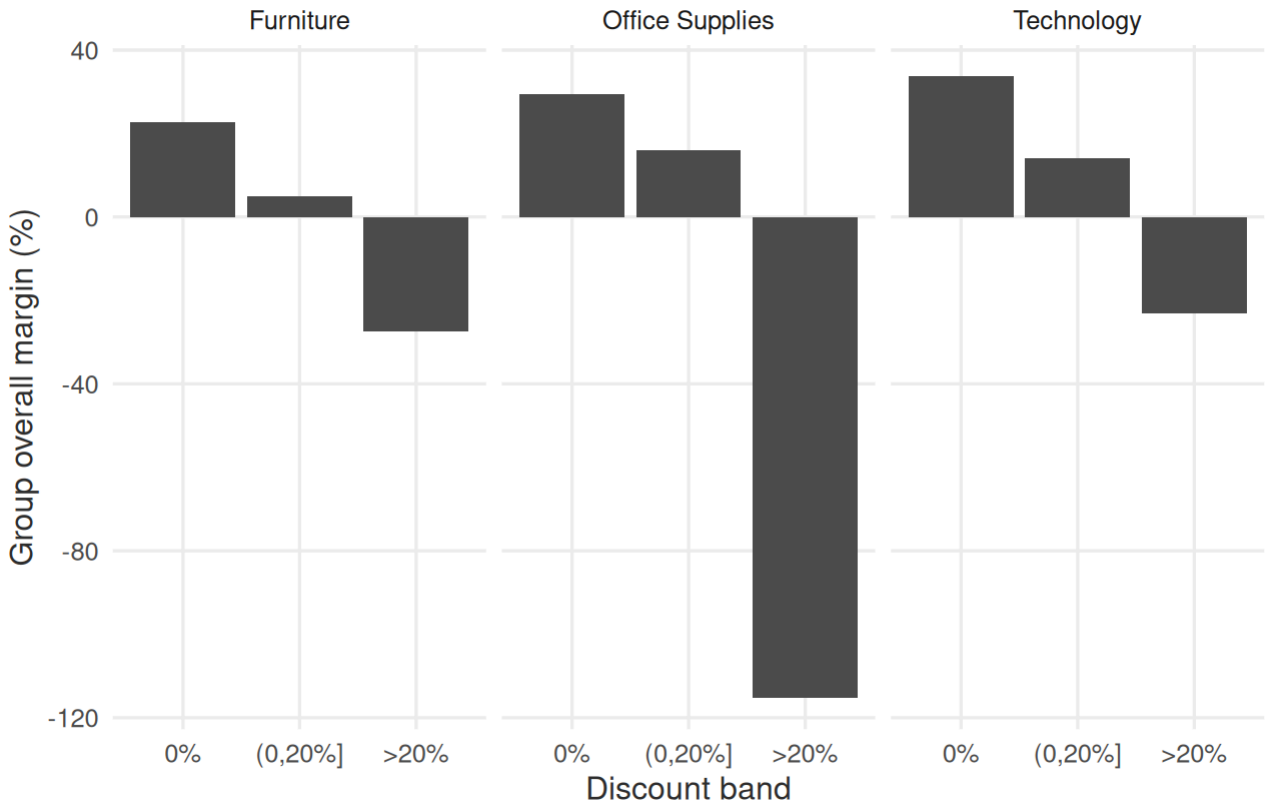
- Compare medians, spread, group sizes and weighted margins.
- The chart gives each line equal weight; the overall margin table does not.
- A negative line margin is possible and is not automatically a bad import.
- Discounts, prices and profit are connected accounting quantities; an association is expected to reflect more than customer response alone.

Keep product mix visible

```
within_category <- lines %>% group_by(category, discount_band) %>%
  summarise(line_count = n(), margin_pct = 100 * sum(profit) / sum(sales),
            .groups = "drop")
knitr::kable(within_category, digits = 2)
```

category	discount_band	line_count	margin_pct
Furniture	None	562	22.80
Furniture	Above 0 to 20%	498	5.05
Furniture	Above 20%	375	-27.45
Office Supplies	None	2096	29.53
Office Supplies	Above 0 to 20%	1480	15.89
Office Supplies	Above 20%	448	-115.25
Technology	None	550	33.73
Technology	Above 0 to 20%	558	13.97
Technology	Above 20%	115	-23.10

```
ggplot(within_category, aes(discount_band, margin_pct)) +
  geom_col(fill = petra_ink) + facet_wrap(~ category) +
  scale_x_discrete(labels = c("None" = "0%", "Above 0 to 20%" = "(0,20%]",
                              "Above 20%" = ">20%")) +
  labs(x = "Discount band", y = "Group overall margin (%)")
```

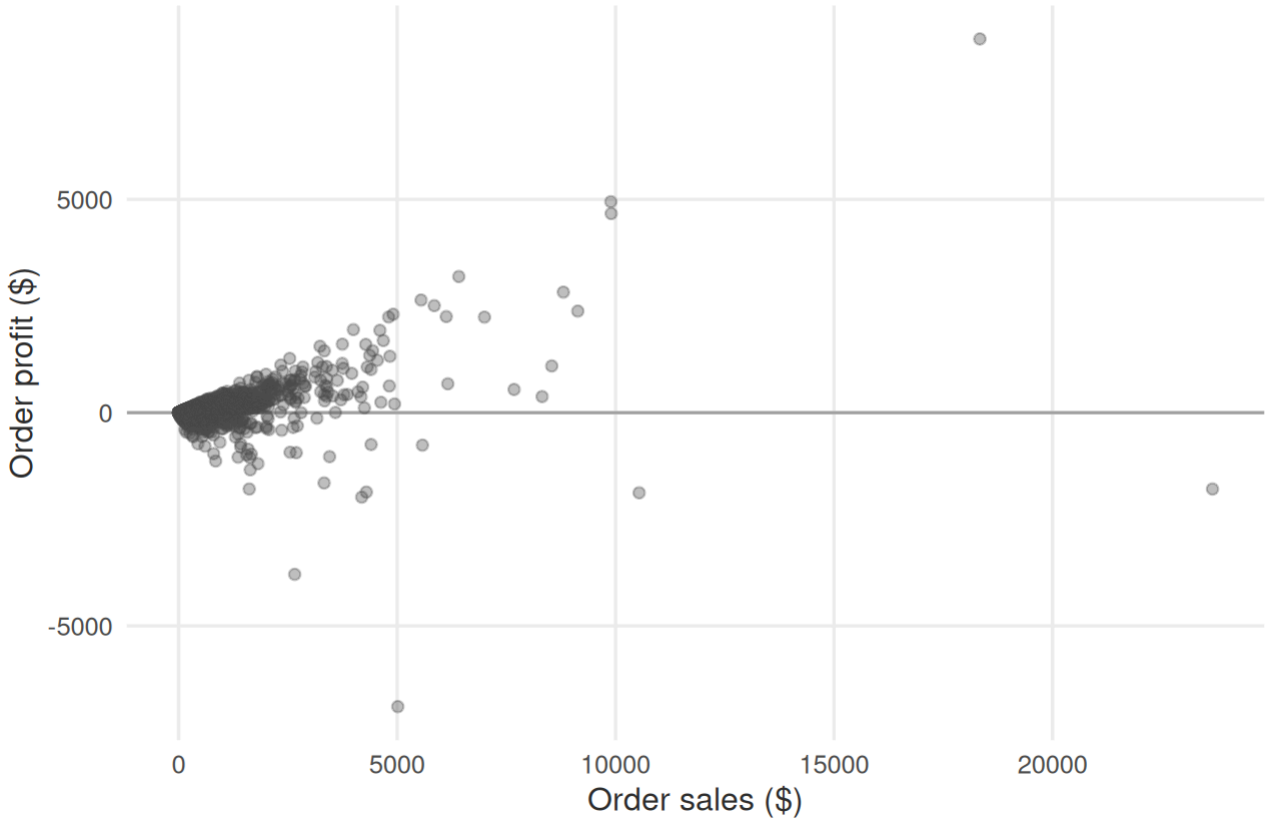


- Stratification compares discount bands within each category.
- It reduces one source of mixing, but does not remove all confounding.
- It does not establish what would happen if the manager changed discounts.

5. Association at order level

> Are higher-sales orders also higher-profit orders?

```
ggplot(orders, aes(sales, profit)) +
  geom_hline(yintercept = 0, colour = "#A3A3A3") +
  geom_point(alpha = .35, colour = petra_ink) +
  labs(x = "Order sales ($)", y = "Order profit ($)")
```




```
correlations <- orders %>% group_by(segment) %>%
  summarise(orders = n(), pearson_r = cor(sales, profit),
            spearman_r = cor(sales, profit, method = "spearman"), .groups = "drop")
knitr::kable(correlations, digits = 3)
```

segment	orders	pearson_r	spearman_r
Consumer	1710	0.364	0.495
Corporate	1021	0.757	0.505
Home Office	591	0.221	0.576

- Pearson measures linear association; Spearman uses ranks.
- Large observations can strongly influence Pearson correlation.
- Segment correlations can differ from the overall correlation.
- Correlation does not show the distribution of losses or establish a causal effect.
- Repeated customers matter if we later try to model orders as independent future observations.

6. Turn a pattern into a decision question

Supported wording:

- “In this extract, the highest discount band has weaker recorded profitability.”
- “Furniture’s overall margin warrants a closer subcategory review.”

Before proposing a discount change, obtain:

- Product cost and accounting definitions.
- Promotion timing and reasons for discount assignment.

- Comparable product groups and customer demand evidence.
- A planned experiment or another defensible causal design.
- Do not claim that removing discounts would reproduce the undiscounted group's margin.
- Do not describe a line-count loss rate as the percentage of customers losing money.

7. Independent commercial briefing

1. Rank subcategories by total profit and by overall margin. Include sales and line counts.
2. Select one weak-performing subcategory and compare its discount bands.
3. Find its three largest loss-making lines; retain them in the main analysis.
4. Compare order-level loss percentages across customer segments.
5. Produce two charts answering different questions.
6. Write a recommendation separating an observed finding, a plausible explanation and evidence still needed.

```
# Use lines for product/discount questions and orders for basket questions.
# A group loss percentage is 100 * mean(profit < 0).
# Calculate rates before overwriting any column with a group total.
```

Submission:

- One compact results table.
- Two labelled charts with units and denominators.
- Five short briefing bullets, including one limitation and one proposed follow-up.

8. Move from description to inference

- Part 3 covers probability, estimation, hypothesis testing and regression.
- The descriptive evidence here concerns the complete fixed extract; p-values are not required to describe its totals.
- `data/business/README.md`.